FIAP GRADUAÇÃO

2TSCPR

Data Science & Machine Learning

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2025 - 1.o Semestre
07/08/2025

Introdução a Ciência de Dados

Agenda

Apresentação
Ciência de Dados
Cientista de Dados
Hello World

Quem sou eu na fila do pão

Marcelo Peres, Cientista de Dados Sênior na **Pravaler**(fintech de educação).
Professor de Graduação na **FIAP**.
Certificado pela **Microsoft** DP-100.
Ex-aluno FIAP de Análise de Sistemas.
Pai da Bia(14 anos), skatista, corredor, gamer
Fui consultor de Advanced Analytics da **IBM** (Vale) e fiz parte do time de Customer Analytics no **WillBank**.

Era Digital

Vaticano em 2005,
Conclave do Papa Benedict XVI

Era Pós-Digital

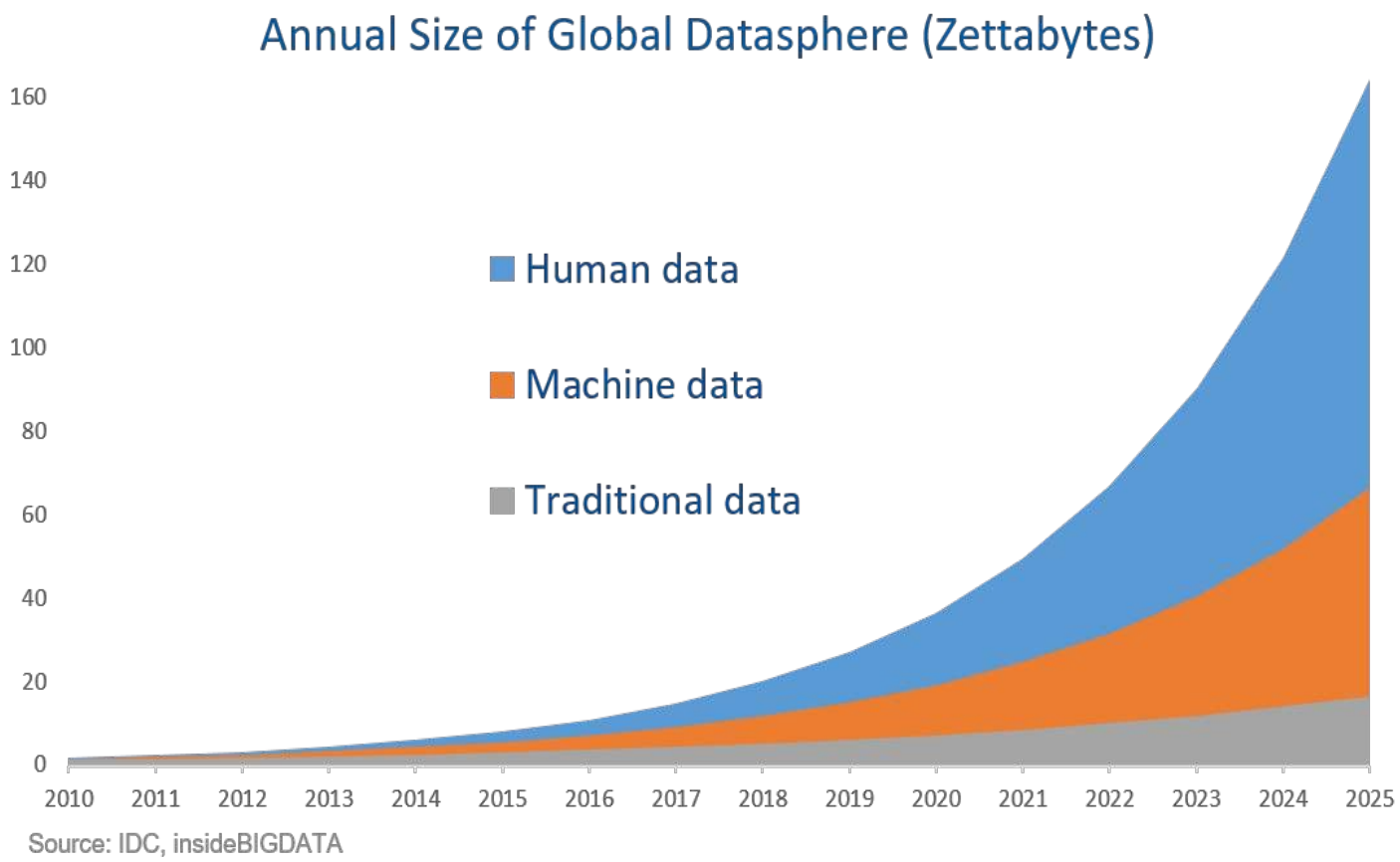
Vaticano em 2013,
Conclave do Papa Francisco

SPONSOR CONTENT JORIS TOONDERS, YONEGO

DATA IS THE NEW OIL OF THE DIGITAL ECONOMY

WIRED,
2014

Data Flair
Data Science Use Cases
facebook
Social Analytics
amazon
Improving E-commerce experience
UBER
Optimizing Rides
Bank of America
Increasing Customer experience
Airbnb
Improving Searches
Spotify
Music Recommendation


Annual Size of Global Datasphere (Zettabytes)
160
140
120
100
80
60
40
20
0
Human data
Machine data
Traditional data
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Source: IDC, insideBIGDATA

1.

“Vivemos em um mundo com excesso de informação. A enorme quantidade de dados aos quais as pessoas são expostas diariamente é prejudicial em todos os sentidos”

2.

“Essas novidades causarão problemas sérios: as pessoas perderão sua capacidade de memória por falta de uso”

3.

“É um perigo obter informações por este meio. As pessoas estão ficando cada vez mais isoladas, não sabem nada sobre como as notícias estão sendo produzidas”

4.

“Essa novidade está acabando com nossas crianças, atolando seus cérebros com informações complexas e desnecessárias, além de prejudicar seu desenvolvimento físico por causa do sedentarismo”

1.
CONRAD GESSNER, cientista suíço no século XVI, sobre a prensa mecânica

2.
SÓCRATES, o “pai da Democracia”, sobre a escrita

3.
MALESHERBES, estadista Francês no séc. XVIII, sobre a novidade da época, os jornais impressos

4.
Conclusão de um artigo publicado no periódico **“SANITARIAN”, EM 1883**, sobre as escolas

The Sanitary News.

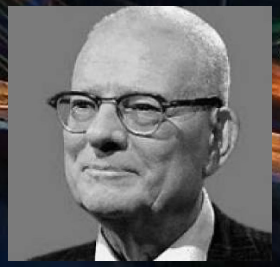
JOURNAL OF SANITARY SCIENCE.

VOLUME VII.

O QUE NÃO É MEDIDO NÃO É GERENCIADO

W. EDWARD DEMING

GURU DA QUALIDADE



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TUDO
PODE SER MEDIDO

DOUGLAS W. HUBBARD

AUTOR DE COMO MEDIR QUALQUER COISA

THE PERFECT HANDSHAKE

CHEVROLET

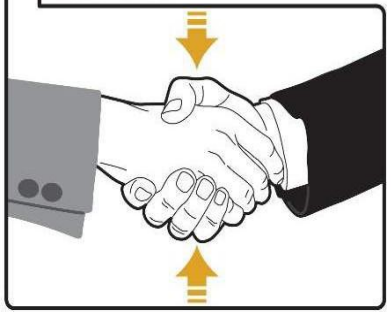
$$PH = \sqrt{(e^2 + ve^2)(d^2) + (cg + dr)^2 + \pi\{(4<s>2)(4<p>2)\}^2 + (vi + t + te)^2 + \{(4<c>2)(4<du>2)\}^2}$$

(e) is eye contact (1=none; 5=direct) 5; (ve) is verbal greeting (1=totally inappropriate; 5=totally appropriate) 5; (d) is Duchenne smile - smiling in eyes and mouth, plus symmetry on both sides of face, and slower offset (1=totally non-Duchenne smile (false smile); 5=totally Duchenne) 5; (cg) completeness of grip (1=very incomplete; 5=full) 5; (dr) is dryness of hand (1=damp; 5=dry) 4; (s) is strength (1= weak; 5=strong) 3; (p) is position of hand (1=back towards own body; 5=other person's bodily zone) 3; (vi) is vigour (1=too low/too high; 5=mid) 3; (t) is temperature of hands (1=too cold/too hot; 5=mid) 3; (te) is texture of hands (5=mid; 1=too rough/too smooth) 3; (c) is control (1=low; 5=high) 3; (du) is duration (1= brief; 5=long) 3

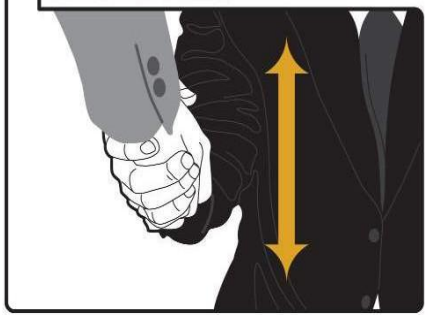
1 RIGHT HAND

2 DRY PALM

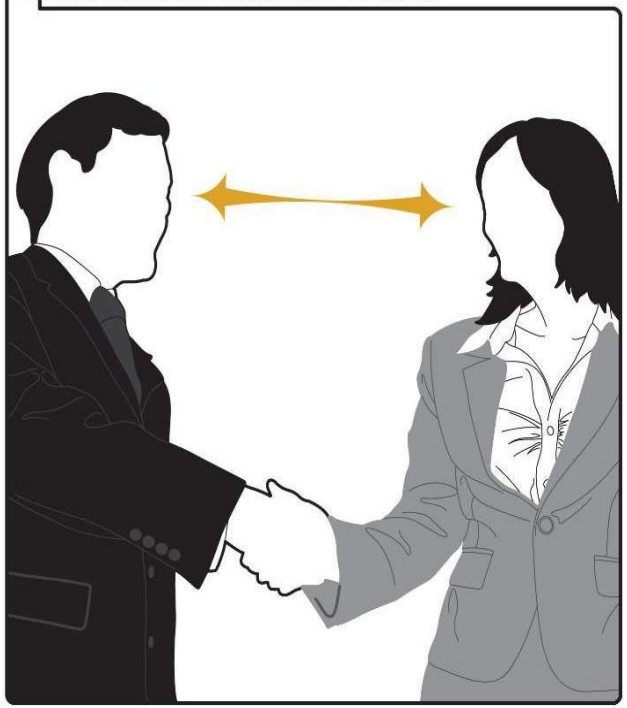
3 STRONG GRIP WITH FINGERS UNDER RECEIVING PALM

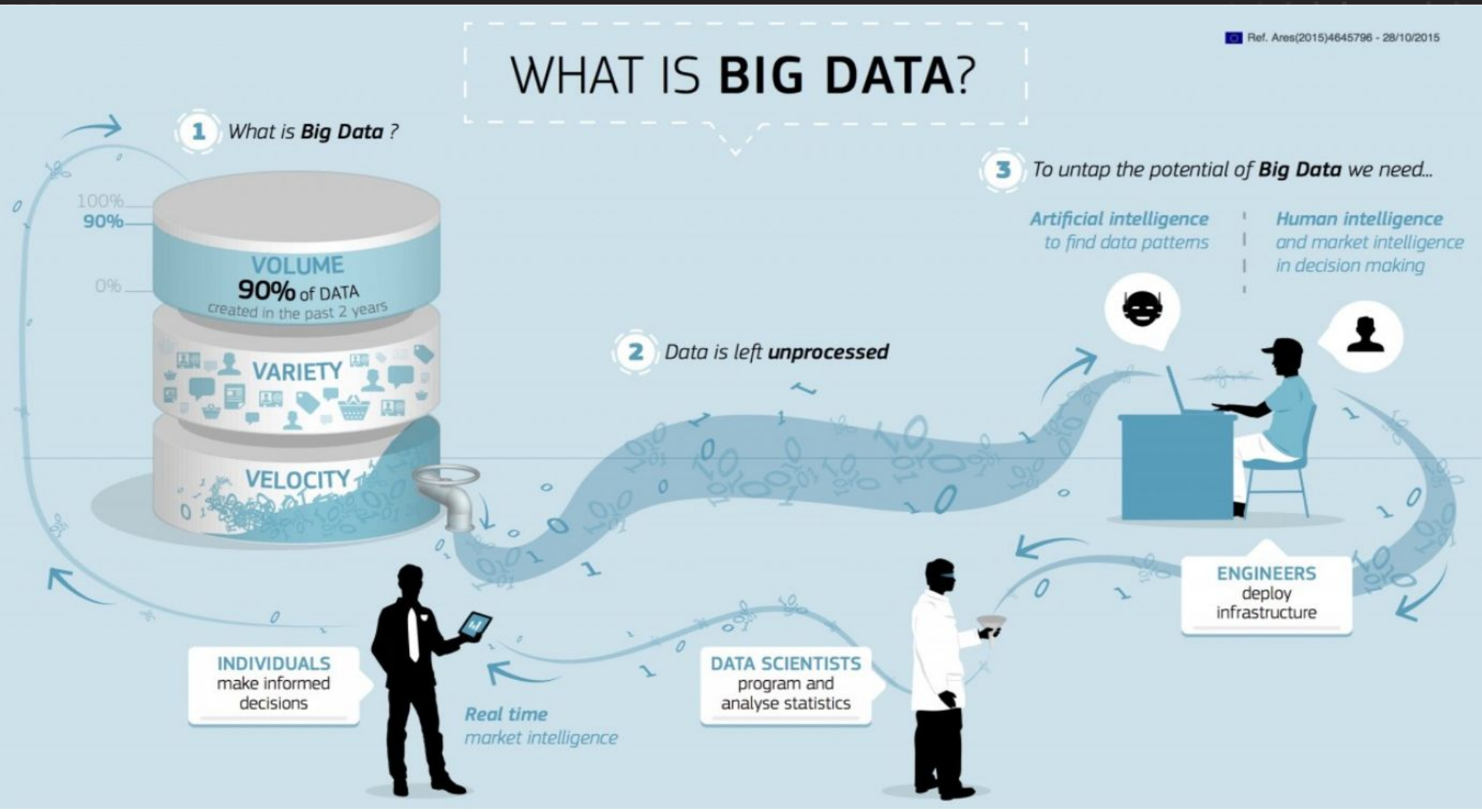


4 THREE TO FOUR VIGOROUS SHAKES FOR TWO TO THREE SECONDS

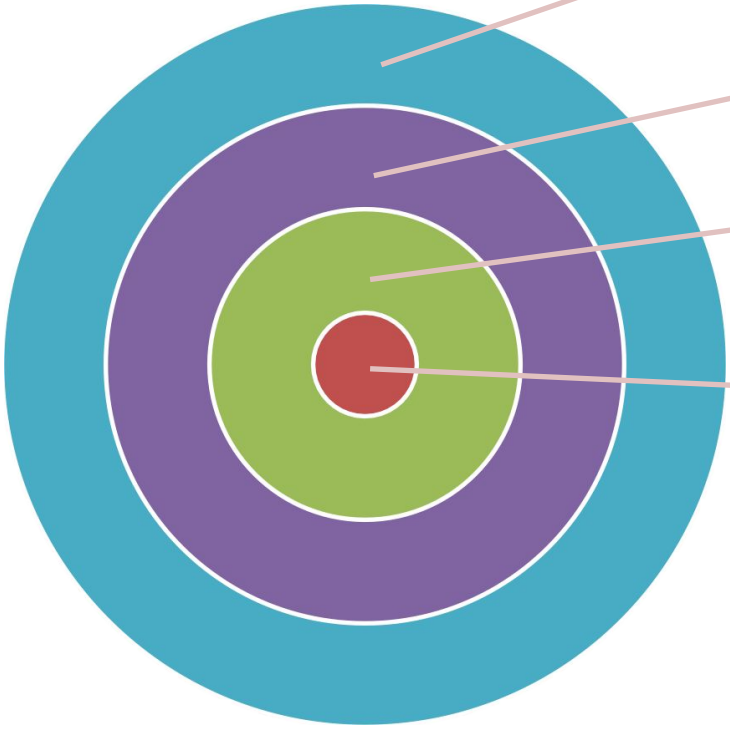


5 EYE CONTACT AND APPROPRIATE SMILE THROUGHOUT




Ref. Ares(2015)4645796 - 28/10/2015
WHAT IS BIG DATA?
1 What is Big Data ?
100%
90%
0%
VOLUME
90% of DATA
created in the past 2 years
VARIETY
VELOCITY
2 Data is left unprocessed
3 To untap the potential of Big Data we need...
Artificial intelligence
to find data patterns
Human intelligence
and market intelligence
in decision making
ENGINEERS
deploy
infrastructure
INDIVIDUALS
make informed
decisions
Real time
market intelligence
DATA SCIENTISTS
program and
analyse statistics

CIÊNCIA DE DADOS



Análise descritiva
O que aconteceu?

Análise inferencial
Como aconteceu?

Análise preditiva
O que acontecerá agora?

Análise prescritiva
Como devo reagir?

CIÊNCIA DE DADOS

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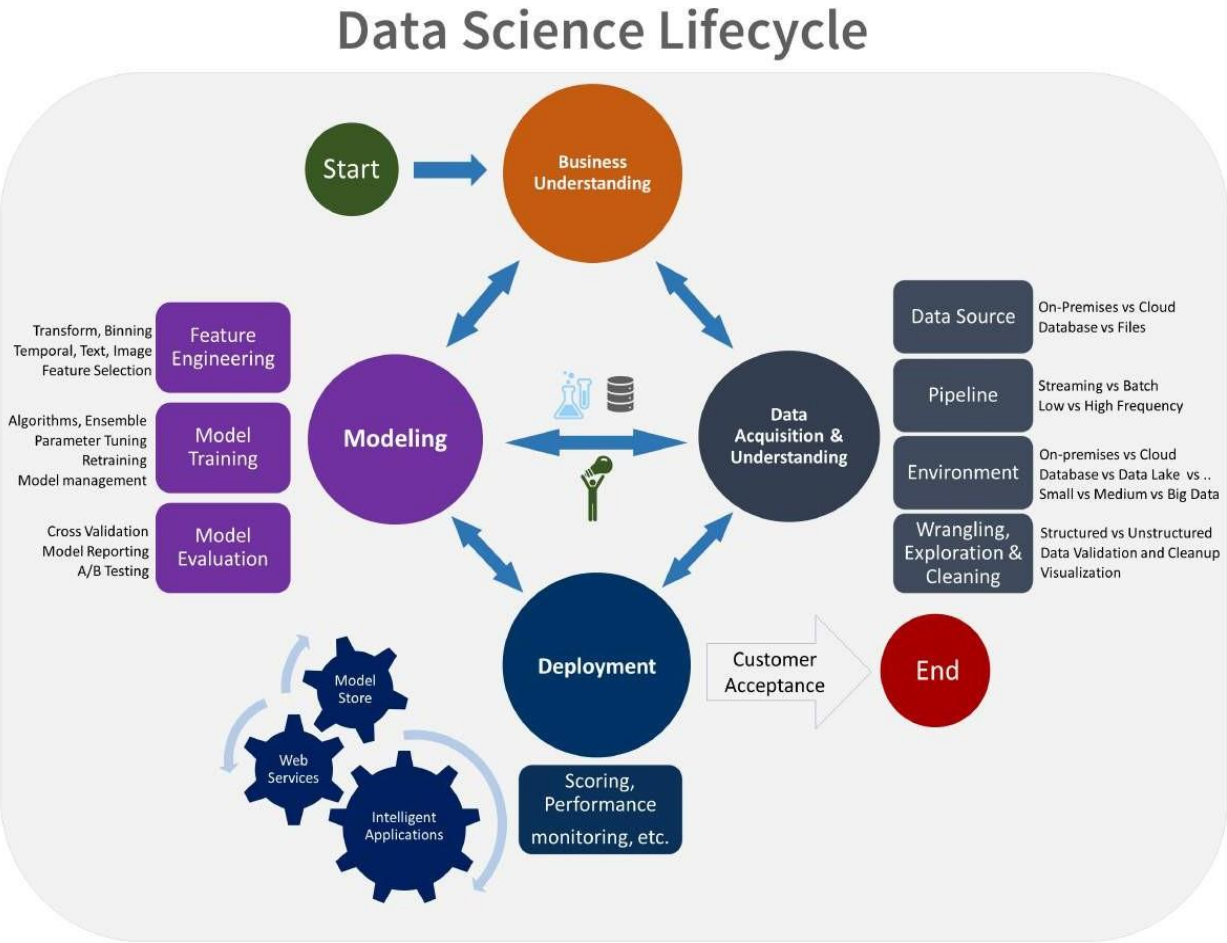
CIÊNCIA DE DADOS



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ARTIFICIAL INTELLIGENCE

IS NOT NEW

ARTIFICIAL INTELLIGENCE

Any technique which enables computers to mimic human behavior

MACHINE LEARNING

AI techniques that give computers the ability to learn without being explicitly programmed to do so

DEEP LEARNING

A subset of ML which make the computation of multi-layer neural networks feasible

1950's 1960's 1970's 1980's 1990's 2000's 2010s

ORACLE

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CIÊNCIA DE DADOS

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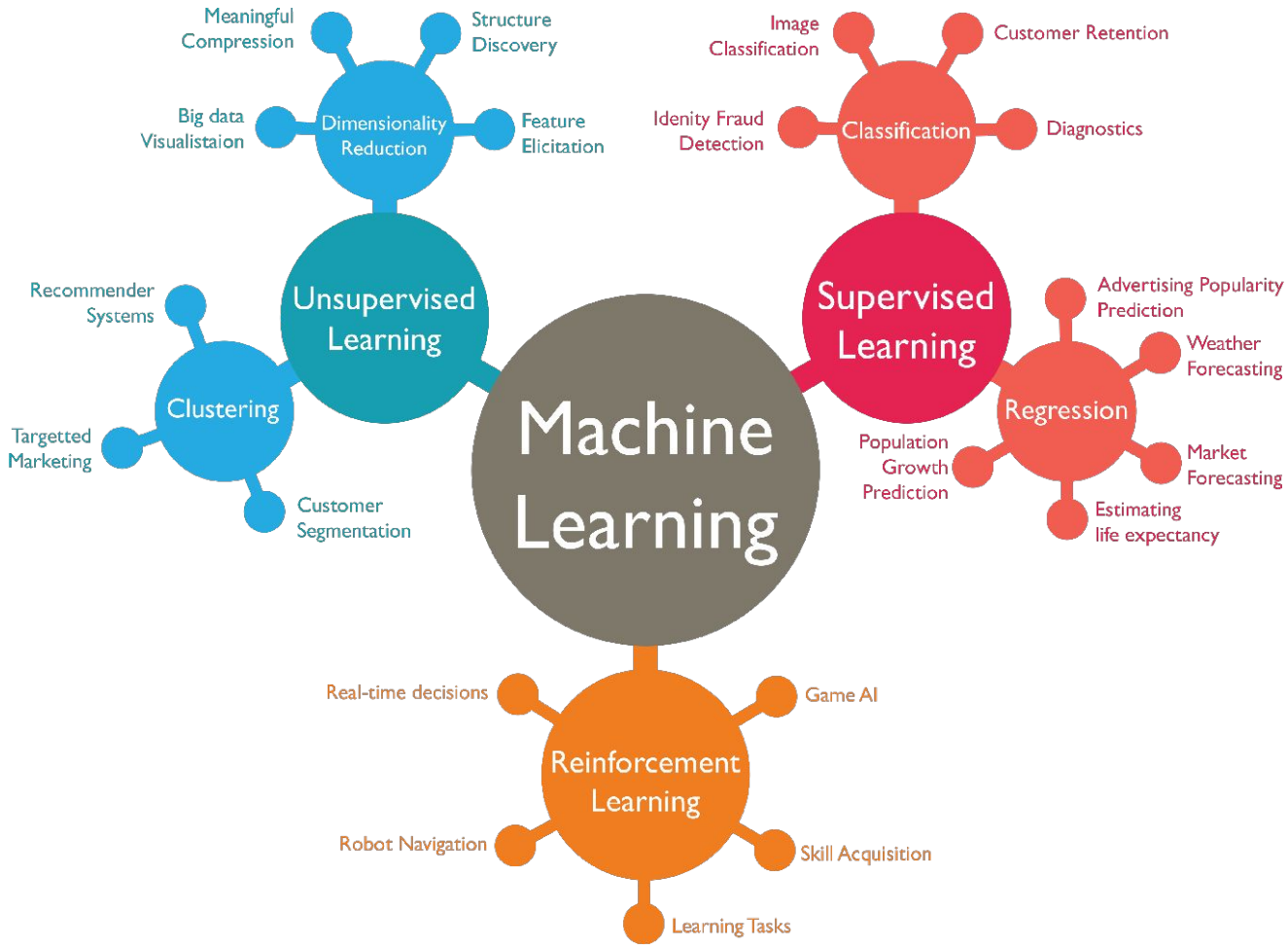
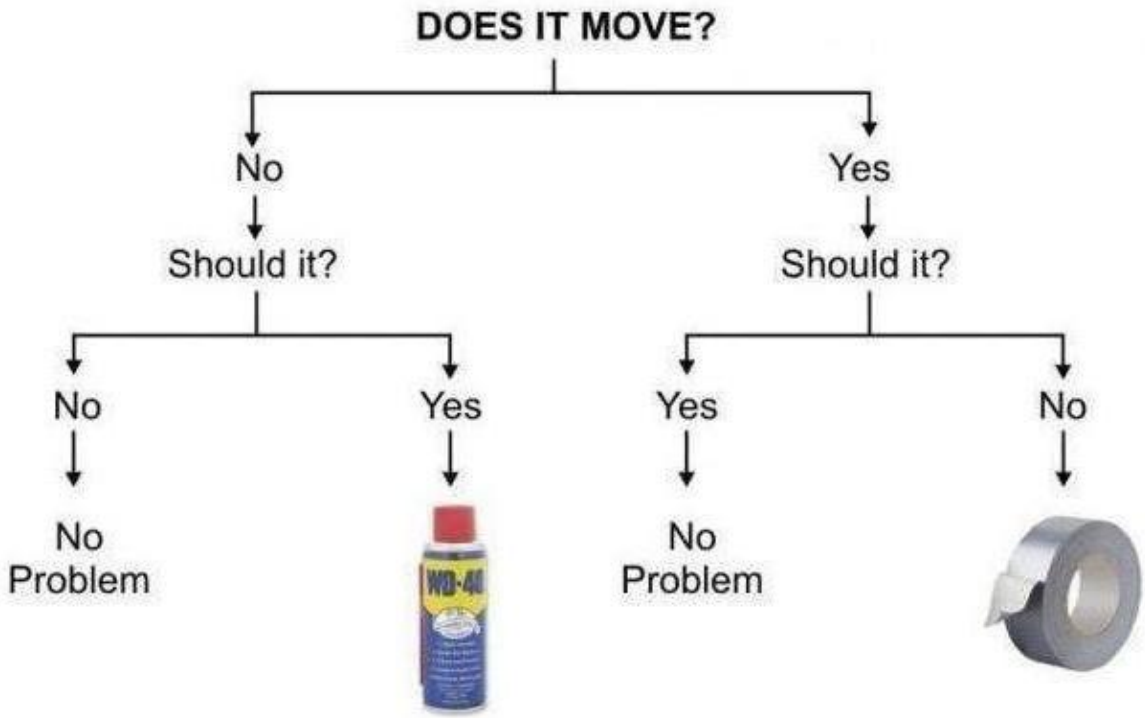


Árvore de decisão



CIÊNCIA DE DADOS

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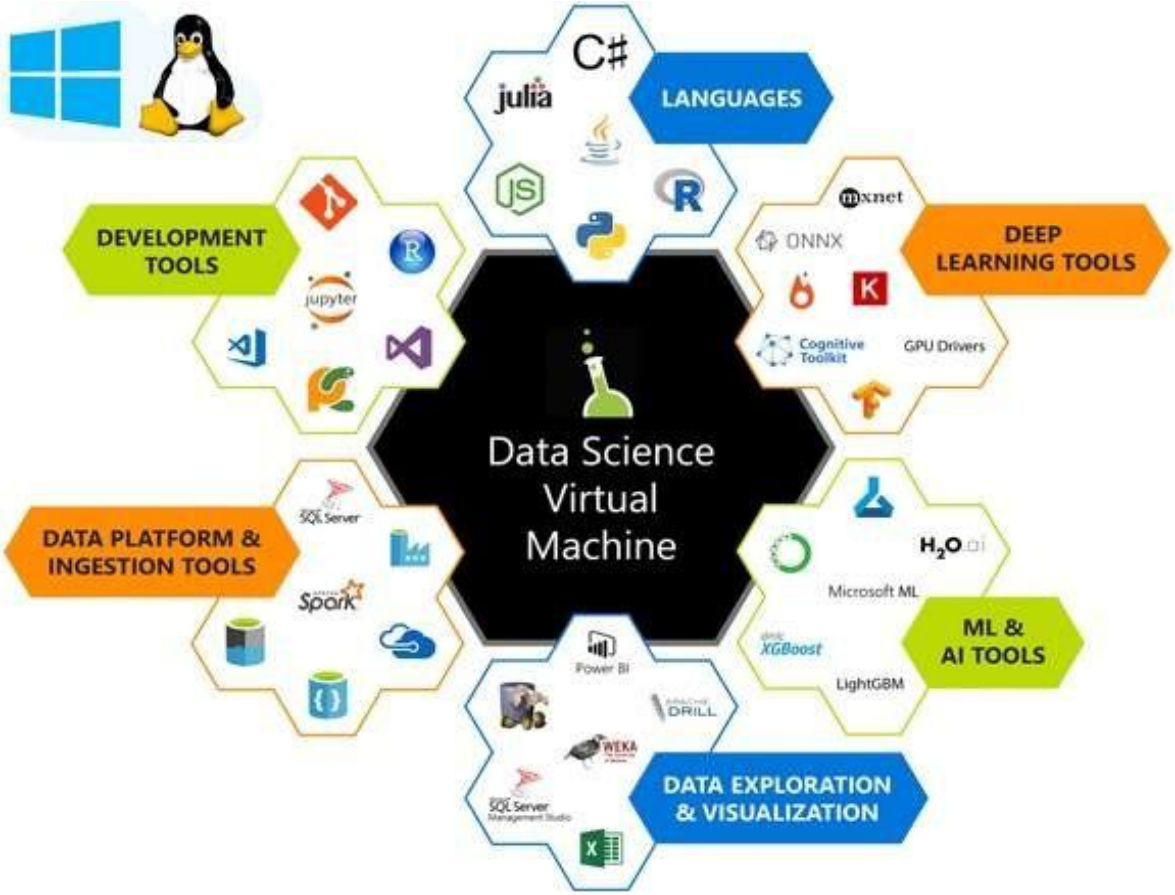

CIENTISTA DE DADOS

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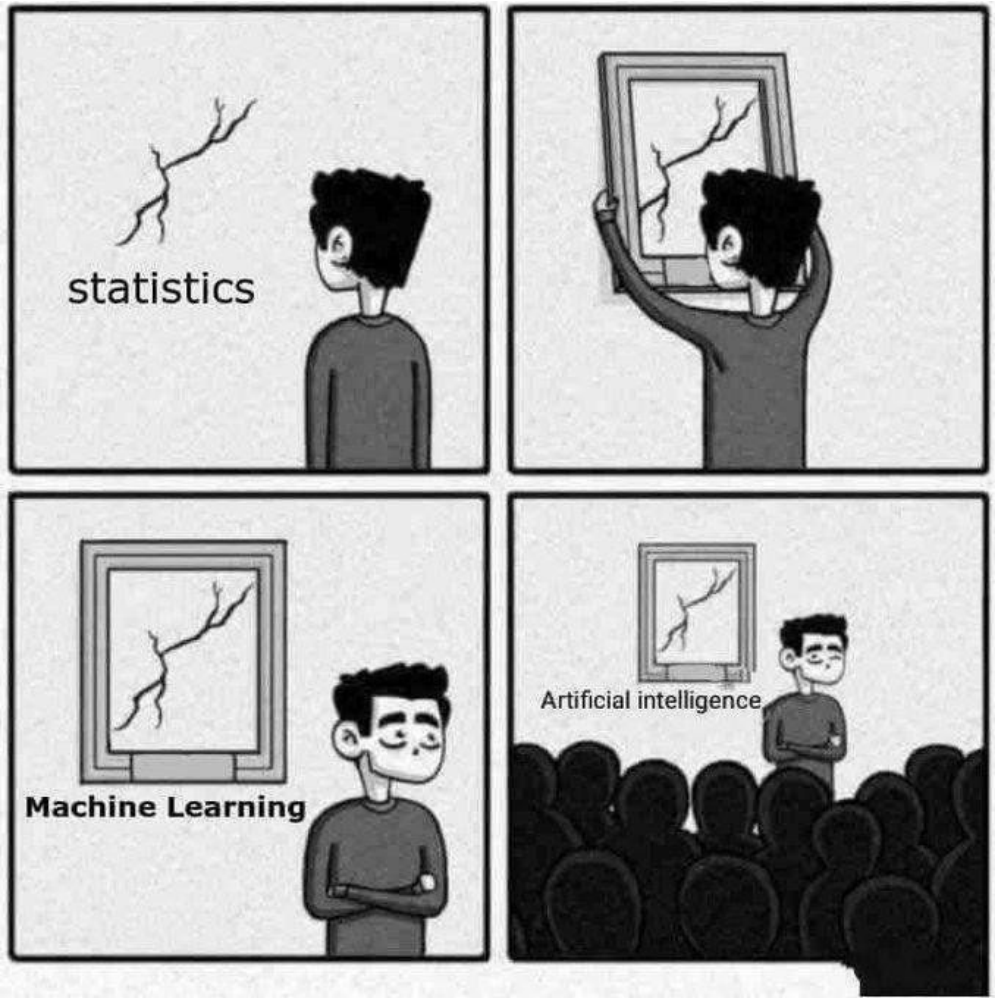
DATA

Data Scientist: The Sexiest Job of the 21st Century

by Thomas H. Davenport and D.J. Patil

CIENTISTA DE DADOS



MODERN DATA SCIENTIST

Data Scientist, the sexiest job of 21th century requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- Machine learning
- Statistical modeling
- Experiment design
- Bayesian inference
- Supervised learning: decision trees, random forests, logistic regression
- Unsupervised learning: clustering, dimensionality reduction
- Optimization: gradient descent and variants

PROGRAMMING & DATABASE

- Computer science fundamentals
- Scripting language e.g. Python
- Statistical computing package e.g. R
- Databases SQL and NoSQL
- Relational algebra
- Parallel databases and parallel query processing
- MapReduce concepts
- Hadoop and Hive/Pig
- Custom reducers
- Experience with xaaS like AWS

DOMAIN KNOWLEDGE & SOFT SKILLS

- Passionate about the business
- Curious about data
- Influence without authority
- Hacker mindset
- Problem solver
- Strategic, proactive, creative, innovative and collaborative

COMMUNICATION & VISUALIZATION

- Able to engage with senior management
- Story telling skills
- Translate data-driven insights into decisions and actions
- Visual art design
- R packages like ggplot or lattice
- Knowledge of any of visualization tools e.g. Flare, D3.js, Tableau

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Entrar no Kaggle.com para rodar um projeto Hello World de Machine Learning

https://www.kaggle.com/code/brendan45774/titanic-top-solution

TSC
Tecnologia em Ciência de Dados

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